The Repression Wedge ω

What it is, how it is “measured” by FTPL inversion, and how it reconciles with the CCL replication

Companion note to §4.1.5 of the dissertation · generated from make_wedge_note.py · all figures recomputed from the 2021–2024 clean window

1 The one-paragraph answer

ω is not a free knob and it is not read off two observed prices. It is recovered by inverting the government’s steady-state backing identity. We observe how much debt the UK carries (v), how small its primary balance is (s), and what fraction of the debt is the captive long segment (b^L). Solvency requires that debt be backed each period by (1−β)v. The primary balance covers only part of that — in every clean-window year it is in deficit — so a residual repression rent R = (1−β)v − s must make up the difference. That rent, expressed per unit of captive long-bond value, maps one-for-one to a price wedge ω through the model’s pricing relation. Averaged over 2021–2024 the recovered wedge is $\bar{\omega}$ = 0.139 ≈ 0.14 (band [0.12, 0.21]). Converted to a per-period convenience-yield spread via the bond’s duration it is of order 94–154 bps/year — the same order of magnitude as the realized-return wedge from the CCL replication (≈ 50–250 bps/yr).

2 Two distinct objects both called “the wedge”

Much of the confusion comes from one word covering two different quantities, in different units, recovered by different procedures. Keep them apart:

Aspect	Model wedge ω (this note)	CCL realized-return wedge
What	Price-level gap on captive long gilts: $1+\omega = Q^L/Q^{L,f}$	Realized return on assets minus the kernel/benchmark return on the surplus claim
Unit	Dimensionless stock (a price ratio) ≈ 14%	Per-period flow return spread, %/yr
How recovered	Invert the FTPL backing identity from v, s, b^L (calibrate.py)	Realized BS returns vs an asset-pricing benchmark needing β_S (ccl_uk/)
Value	$\bar{\omega}$ = 0.139, band [0.12, 0.21]	≈ −0.52%/yr (empirical β_S) to −2.49%/yr (Damodaran)
Role	Primitive that pins down the model steady state	Independent empirical cross-check

This note is about the left column — the model primitive — and in particular about the sentence in §4.1.5 that ω is “recovered from prices rather than assigned.” The remainder explains precisely what that recovery is.

3 The structural definition (and why we cannot read it directly)

Structurally the wedge is the proportional gap between the price the captive sector pays for the long gilt and the price an unconstrained investor would pay for the same cash flows, discounting with the economy’s stochastic discount factor:

$$1 + \omega = Q^L / Q^{L,f}$$

The fundamental price $Q^{L,f} = β/(\bar{π} − βδ)$ is a model object: no unconstrained, frictionless long-gilt price is observed in the UK data, because the long end is the captive segment. So we cannot form the ratio directly. Instead we recover ω from the one place the captive distortion leaves an observable footprint: the government's books.

4 The FTPL inversion, step by step

4.1 The steady-state backing identity

In steady state the model's valuation equation reduces to a single accounting identity: the per-period real flow that services the debt must equal the net discount times the debt's real value. That flow is split between the conventional primary balance and the repression rent:

$$s/v + R/v = 1 − β$$

Read it plainly: carrying debt worth v (as a share of GDP) costs (1−β)v per period at the riskless rate. Conventional public finance assumes the primary balance s covers that. In the UK it does not — s is negative every year — so something else must back the debt. The model says it is the rent extracted from captive holders. Rearranging gives the rent as a residual:

$$R = (1 − β)\cdot v − s$$

4.2 From rent to wedge

The rent is generated by holding the captive long bonds above fundamental value. Expressed per unit of long-bond market value $b^L_{mkt} = Q^L b^L$, the rent share equals the per-period shadow value of the captive constraint; through the pricing relation that share inverts to the price wedge:

$$\text{rent\_share} = R / b^L_{mkt} \quad \Longrightarrow \quad ω = \text{rent\_share} / (1 − \text{rent\_share})$$

That last step is the algebra in calibrate.py lines 88–96. The logic is purely an accounting-plus-pricing inversion: data on (v, s, b^L) → required rent R → rent share → ω. Nothing is fit by regression and no return series is differenced; ω is whatever makes the observed debt exactly backed under the model's pricing of the captive bond. This is what "measured by inversion" means.

5 Worked example: 2024

Step	Expression	2024 value
Total gilt MV / GDP	v	0.5801
Primary balance / GDP	s	-2.374% (deficit)
Long MV / GDP	$b^L_{mkt} = 0.3532\cdot v$	0.2049
Required backing	$(1−β)\cdot v$	0.0058 (0.58% GDP)
Residual rent	$R = (1−β)v − s$	0.0295 (2.95% GDP)
Rent share	R / b^L_{mkt}	0.1442
Wedge	ω = rent_share/(1−rent_share)	0.1685

The intuition jumps out of the arithmetic: because the 2024 primary balance is a 2.37%-of-GDP deficit, it does not just fail to back the debt — it adds to the burden. The rent must therefore supply the entire required backing plus the deficit (0.58% + 2.37% = 2.95%

of GDP). A large rent on a long segment worth only ~20% of GDP implies a large price wedge, ω = 0.168. Years with smaller deficits (e.g. 2022) back out smaller wedges (ω = 0.092).

6 Clean-window results, 2021–2024

Year	v	s	R (resid.)	rent share	ω	ω/D at DMO dur.
2021	0.753	-2.98%	0.037	0.140	0.163	142 bps
2022	0.599	-1.18%	0.018	0.084	0.091	100 bps
2023	0.586	-1.87%	0.025	0.119	0.134	160 bps
2024	0.580	-2.37%	0.030	0.144	0.168	215 bps
Avg					0.139	154 bps

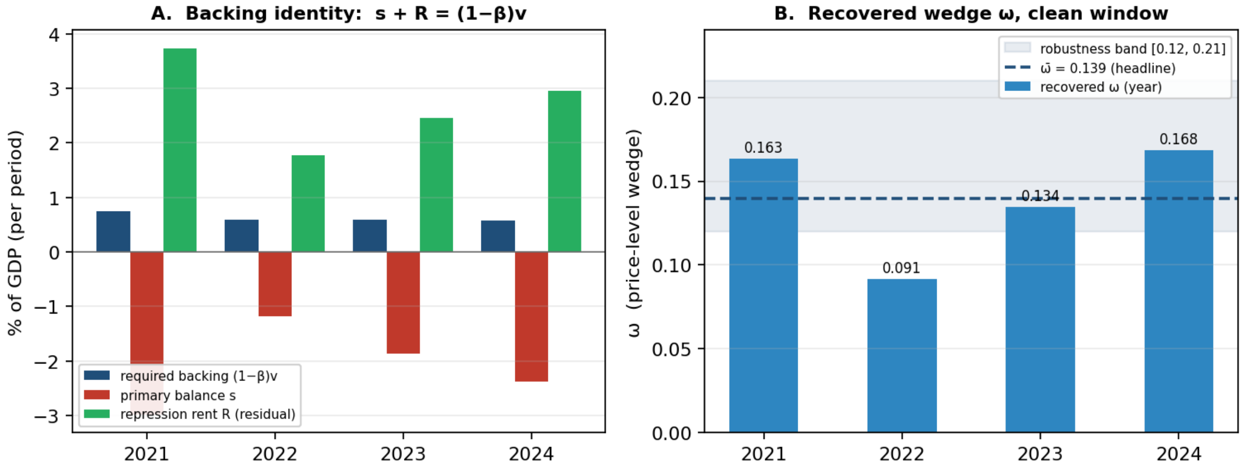

Figure 1. (A) Each year the required backing (1−β)v is small, the primary balance s is negative, and the residual rent R closes the gap. (B) The recovered wedge ω by year, the four-year headline ω̄ = 0.139, and the [0.12, 0.21] robustness band driven mainly by the long-share split.

7 From a price stock to a yield flow: the duration bridge

ω is a stock — a one-off gap in the price level of the bond. The financial-repression literature, and the CCL replication, usually speak in flow terms: a per-period yield or return spread. The two are linked, to first order, by the bond's duration D: a price that sits ω above fundamental is equivalent to paying a yield D×spread lower each period, so

$$\text{spread} \approx \omega / D \quad \Longleftrightarrow \quad \omega \approx D \cdot \text{spread}$$

The implied spread depends on which duration you divide by, and the calibration contains two. (i) The structural model uses a single δ-implied steady-state duration of D ≈ 14.8 years; with $\bar{\omega}$ = 0.139 this gives $\bar{\omega}$/D ≈ 94 bps/yr, consistent with the draft's exact shadow-price figure $\tilde{\nu}$ ≈ 83 bps/yr (§4.1.5). (ii) The empirical DMO portfolio durations actually used in the inversion are shorter (≈ 8–11 years), which raises the per-year bridge to an average ≈ 154 bps/yr. Either way the convenience-yield equivalent is of order 94–154 bps/yr — squarely inside the range the repression literature documents for advanced-economy sovereigns. The duration discipline is what keeps it credible: had the decay implied a 3-year duration, the same 14% price wedge would require an implausible ~3.9%/yr spread.

8 Reconciliation with the CCL replication

Fair statement. The model wedge is calibrated from data through the valuation identity ($\bar{\omega}$ = 0.139, a price-level gap). Expressed as a per-period convenience-yield spread (~94–154 bps/yr via the bond's duration) it is the same order of magnitude as the realized-return wedge from the CCL replication (≈ 0.5–2.5%/yr), which therefore serves as an independent cross-check rather than the same calculation.

Two caveats to keep attached whenever the agreement is claimed:

• Different procedures. The model number is an FTPL accounting-identity inversion on 2021–2024; the CCL number is a realized-return differential over the full sample requiring an estimated β_S. They corroborate in magnitude; they are not the same object computed twice.

• Sign convention. The CCL wedge is reported negative (repression depresses the realized return relative to benchmark); the model spread is positive (the government borrows below fundamental). Compare magnitudes, and state the convention, before putting the two side by side in print.

9 How ω enters the structural model

Once recovered, ω is handed to the structural model as a primitive (model.py line 53, ω = 0.14). It is not re-derived inside the model; rather it pins down the rest of the deterministic steady state, which is "solved, not chosen":

• the observed long price $Q^L = (1+\omega)Q^{L,f}$ (66.47 vs 58.30);
• the shadow price $\tilde{\nu} = [(\bar{\pi}-\beta\delta)/\bar{\pi}]\cdot\omega/(1+\omega)$ = 0.00207, whose reciprocal $1/\tilde{\nu} \approx 482$ is the stiff eigenvalue governing the dynamics;
• the per-period rent R and the steady-state backing split $s/v + R/v = 1-\beta$.

So the precise chain is: data → (FTPL inversion) → $\bar{\omega}$ = 0.14 → (assigned as a primitive) → the model steady state. The wedge is determined by the data via the identity, not by the other model parameters.

10 A labeling correction worth making

The comment at model.py:53 calls ω the "MEASURED wedge (return differential on captive gilts)," and §4.1.5 says it is "equivalently read from the differential between the realized return … and the kernel-implied return." That phrasing describes the CCL realized-return procedure, which the code does not actually run to produce 0.14 — the number is the FTPL-identity inversion. The two agree in magnitude but are distinct computations. Recommended fix: describe ω as recovered from the valuation identity ($1+\omega = Q^L/Q^{L,f}$, via the rent residual), and cite the realized-return differential as an independent corroborating magnitude, not as the definition.

11 Defensible one-liner for the dissertation

"The wedge is calibrated from data through the valuation identity ($\bar{\omega} = 0.14$, a price-level gap); expressed as a per-period convenience-yield spread (of order 1% per year via the bond's duration) it is the same order of magnitude as the realized-return wedge from the CCL replication (~0.5–2.5%/yr), which serves as an independent cross-check."